Syrian Private University · Faculty of Artificial Intelligence Engineering

SEMESTER PROJECT SEMINAR

Detecting Data Leakage in Cloud Storage with Interpretable Machine Learning

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Academic Year 2026–2027

Research paper comparison

Study	Data	Method	Main result	Link to our project
Harahap & Hasibuan (2025), JISI	Kaggle cloud-storage logs: 49,498 events, 9 attributes	Decision tree, depth 3; no imbalance handling	Accuracy 0.708, leakage recall 0.315	Our baseline: barely beats always “normal” (0.689)
Apicella et al. (2025), AI Review	Survey, no dataset	Taxonomy of ML data leakage	Checklist of safe practices	Used to design our leakage-safe pipeline
Hakim et al. (2025), IEEE Access	Threat reports, 10 ATT&CK tactics; 93:7 imbalance	RF, SVM, LR, XGBoost, NB; SMOTE+ENN	RF best, CV accuracy > 0.99	Resampling before CV may inflate scores
Komka et al. (2026), NJESTI	CERT v6.2 + 4 injected cloud attacks	Isolation Forest, One-Class SVM	Weighted F1: IF 0.92, OCSVM 0.81	Same threat; injected attacks make scores optimistic

What we did vs the paper

	Harahap & Hasibuan (2025)	Our work
Model	Decision tree (entropy, maximum depth 3)	Tuned decision tree; XGBoost gained only +0.013 PR-AUC, so we kept the readable tree
Attributes	Nine attributes, one-hot encoded	Same nine plus external destination and hour (+0.122 PR-AUC)
Class imbalance	Not addressed	Class weights (a missed leak counts about 2.2× more); SMOTEN and random over- and undersampling scored lower
Alert threshold	Default 0.5	$\theta^* = 0.416$: the most precise threshold that still catches at least 80% of leaks, set on training data
Evaluation	Accuracy and recall reported	Stratified 80:20 split before any fitting, leakage-safe 5-fold CV, one final test
Explanation	The tree only	The rule behind every alert, plus attribute importance

01 · THE PROBLEM

Data leaves through people who have access

Insiders and stolen accounts

A valid login can open and send confidential files, so nothing looks wrong at the network edge.

Every action is logged

Cloud storage records who touched which file, when, and where it was sent.

Fixed rules fall short

Rule-based DLP raises many false alarms and misses patterns nobody wrote a rule for.

Our goal: learn leakage patterns from the logs with a decision tree an analyst can read, and measure it honestly.

02 · THE BASE ARTICLE

Harahap & Hasibuan (2025) and its data

MODEL	Decision tree (entropy, maximum depth 3)
INPUT	Nine attributes, one-hot encoding (as stated)
SOURCE	“Detecting Data Leakage in Cloud Storage Using Decision Tree Classification”, JISI, 2025

70.8%
accuracy (reported)

31.5%
leakage recall (reported)

Kaggle “Data Leakage Detection”

49,500
events, one per file action

31.1%
labelled as leakage

15 columns: who acted, how they logged in, what they did, how sensitive the file was, where it went, and when.

03 · WHAT WE FOUND

Is 70.8% accuracy actually good?

68.9%

Always “normal”

A model that never raises an alert already gets this accuracy. The article is only 1.9 points higher.

31.5%

Leakage recall

Fewer than one leak in three is caught; about 69% of leaks are missed.

74.3%

Ceiling with nine attributes

91% of events share their exact attribute combination with events of the other label.

The limit is in the attributes, not in the algorithm.

04 · OUR METHOD

A leakage-safe evaluation pipeline

Data and ceiling

Audit the events and compute the best possible accuracy, Acc*

→

Split 80 : 20

Stratified, before any fitting; 9,900 test events locked away

→

Train and tune

Reference tree, new attributes, 3 models, 4 balancers, threshold θ*: training set only

→

Test once

Final scores, comparison with the article's tree, rules and importance

- ✓ Every fitted step (imputer, encoder, resampler) sits inside a pipeline and is refit in each fold.
- ✓ The same stratified 5-fold split is used for every comparison.
- ✓ The alert threshold comes from out-of-fold scores, never from the test set.

04 · IMPROVEMENTS

What we changed to catch more leaks

1 · Two new attributes

+0.122

PR-AUC, 5-fold CV

External destination: 39.0% leakage vs 25.9% internal. Hour: 40.4% outside working hours vs 15.8% inside.

2 · Model choice

+0.013

PR-AUC, XGBoost over the tree

Too small to give up readable rules: we keep the tree. Class weights kept; SMOTEN and resampling scored lower.

3 · Alert threshold

θ* = 0.416

instead of the default 0.5

Chosen on training data only: the most precise threshold that still catches at least 80% of leaks.

Baseline first: our rebuild of the article's tree (one-hot, as the article states) reaches accuracy 0.695 and catches 10.1% of leaks on the same test set, below the 0.708 and 0.315 the article reports; every gain is measured against this verifiable baseline.

05 · RESULTS ON THE TEST SET

Our tree catches 8.2× as many leaks

Leaks caught

Article's tree

311

Our tree

2,547

Recall

Article's tree

0.101

Our tree

0.827

MCC

Article's tree

0.127

Our tree

0.429

PR-AUC

Article's tree

0.452

Our tree

0.609

Locked test set: 9,900 events, 3,079 of them leaks, never used in training. Article's tree = our rebuild of its model (entropy, depth 3, one-hot as the article states, threshold 0.5). MCC: 0 = guessing, 1 = perfect.

05 · RESULTS ON THE TEST SET

The price: more alerts to review

	PREDICTED NORMAL	PREDICTED LEAK
Actual normal	**4,339** correctly left alone	**2,482** false alarms
Actual leak	**532** missed leaks	**2,547** leaks caught

51%

of test events raise an alert (article’s tree: 5.7%)

0.507

precision: about one alert in two is a real leak

≈ 0.70

accuracy of both trees: accuracy hides the difference that matters

06 · EXPLAINING EACH ALERT

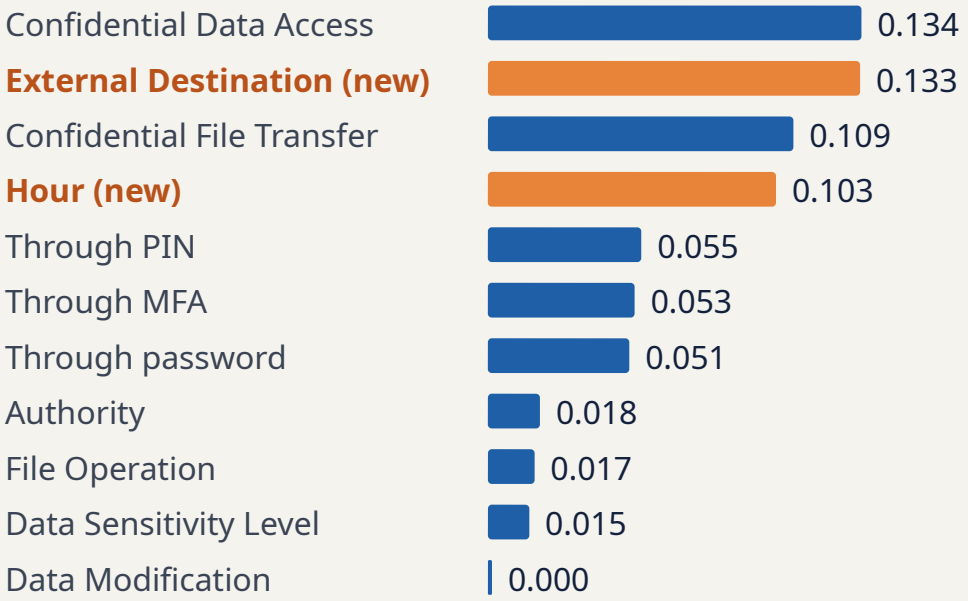
Every alert comes with its reason

STRONGEST RULE IN THE FINAL TREE

```
IF confidential file transfer = yes
AND destination = external
AND authority in {intern, senior manager, staff}
AND password login = no
AND sensitivity in {high, medium}
AND hour between 00:00 and 02:59
THEN leakage
```

369 training events match this path; 61% of them are leaks.

Attribute importance: drop in PR-AUC when the attribute is shuffled (test set)



07 · LIMITS AND NEXT STEPS

What still limits us, and what comes next

Limits

Coarse attributes: 85% of events share their attribute combination with the other label.

Many alerts: about half of all events raise one; one in two is a real leak.

Long rules: 545 leaves; the strongest rules need six to nine conditions.

No user history: each event is judged alone, and the data is synthetic.

Next steps

1 · Evasion test: change leakage events the way an attacker would (working hours, internal destination) and measure the recall drop.

2 · Prototype: a scoring service and a dashboard that return the score, the label and the rule that fired.

3 · Report: add the evasion and prototype results to Chapter 2, then write the abstract and the conclusion.